

Brilliance Base EMS

Complete Role Documentation

Candidate · Employee · Manager · HR

Final Year Project · React Native + Node.js / Express / MongoDB

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This document describes every major module delivered in Brilliance EMS for the four primary operational roles: Candidate, Employee, Manager, and HR. For each role it covers navigation, end-to-end user flows, dynamic (API-backed) features, backend endpoints, data models, and known gaps.

API base path is /api. The mobile app uses configurable API_BASE_URL. Authentication is JWT Bearer token for all protected routes.

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1. System Overview & Shared Foundation

1.1 Architecture

- Frontend: React Native (Android/iOS) — Brilliance/
- Backend: Node.js + Express + MongoDB — Brilliance Backend/
- Auth: JWT (Bearer), optional 2FA, forgot/reset password OTP
- Roles (API / UI): candidate, employee, manager, hr, admin
- Uploads served at /uploads (resumes, avatars, documents, payslip PDFs)

1.2 Shared Auth APIs

POST /auth/register — Candidate (and Admin) public signup
POST /auth/login — All roles
POST /auth/verify-2fa — If 2FA enabled
GET /auth/me — Session refresh
POST /auth/logout
POST /auth/change-password
PATCH /auth/2fa — Enable/disable 2FA
POST /auth/forgot-password | verify-otp | reset-password

1.3 Shared Staff Profile (Employee / Manager / HR / Admin)

GET|PATCH /employee/profile
PUT|DELETE /employee/profile/avatar
GET|POST|DELETE /employee/documents...
GET /employee/assets — Assigned assets read-only
GET /employee/company-policies
GET|PATCH /employee/profile-completion...

1.4 Navigation Model

App.jsx hosts a root Native Stack. After login, role-based Drawer opens Bottom Tabs. Feature screens (ApplicationReview, MTasks, HRApprovalCenter, Payslips, etc.) are registered on the root stack.

2. Candidate Module — Complete Detail

2.1 Purpose

Candidates discover open jobs, submit applications with CV/cover letter, track application status through the recruitment pipeline, and manage a lightweight personal profile. Candidates cannot self-register as staff; hiring converts them to Employee via HR.

2.2 Navigation

Bottom Tabs (CandidateTabs)

- Home — CHome.jsx
- Jobs — CJob.jsx
- Applications — CApplication.jsx
- Profile — Profile.jsx

Drawer

- Home, Notifications, Settings, Privacy Policy, Terms of Service

Stack Screens

- CandidateSetup.jsx — forced when personal profile incomplete
- JobDetail.jsx, ApplyJob.jsx, ApplicationDetail.jsx
- EditPersonalInfo.jsx, ChangePassword, TwoFactorAuth, Notifications

2.3 User Flows

A. Signup & Setup

1. RoleSelection !' Sign Up as Candidate
2. POST /auth/register creates User with role=candidate
3. If incomplete !' CandidateSetup (personal info + optional avatar)
4. PUT /candidates/me/profile !' main app

B. Browse & Apply

1. Jobs tab: GET /jobs?status=Active (RecruitmentContext)
2. JobDetail !' ApplyJob form (phone, experience, education, salary, LinkedIn, cover letter "e20 chars)
3. Attach resume PDF/DOC !' POST /applications multipart !' status Applied
4. Duplicate apply to same active job blocked (409)

C. Track Applications

- GET /applications/me — list own applications
- Pipeline: Applied !' Review !' Shortlisted !' Interview !' Selected | Rejected | Withdrawn
- Reschedule interview: POST /applications/:id/reschedule
- Withdraw: POST /applications/:id/withdraw

D. Profile

- Avatar upload/remove; edit personal information; password / 2FA in Settings

2.4 Candidate APIs

```
GET /candidates/me/profile
PUT /candidates/me/profile
PUT|DELETE /candidates/me/avatar
GET /jobs - List / filter Active
GET /jobs/:id
POST /applications - Multipart apply
GET /applications/me
GET /applications/:id
POST /applications/:id/reschedule
POST /applications/:id/withdraw
```

2.5 Key Models

User: role candidate + candidateProfile (personalInfo, avatar, isProfileComplete)

Job: title, dept, location, types, salary, requirements, status Active/Closed

Application: job, candidate, coverLetter, resume, status, interview, reviewNote, rescheduleRequest

2.6 Dynamic vs Gaps

- DYNAMIC: jobs, apply, applications list/detail, withdraw, reschedule, profile, avatar, auth
- GAP: Notifications mostly local; Privacy/Terms are static screens

3. Employee Module — Complete Detail

3.1 Purpose

Employees (created by HR hire or Add Employee) use self-service EMS: attendance clock, leave, tasks, overtime, expenses, payslips, loans/advances, performance reviews from managers, company policies, and master profile (personal, emergency, bank, documents, assets).

3.2 Navigation

Bottom Tabs

- Home — EHome.jsx | Attendance — EAttendance.jsx | Leave — ELeave.jsx
- Tasks — ETasks.jsx | Profile — Profile.jsx

Drawer

- My Profile, Inbox, Payslips, Overtime, Expenses, Loan & Advance, Performance, Company Policies

Stack Screens

- ApplyLeave, AttendanceRequest (Correction + WFH)
- Payslips, ExpenseClaim, ApplyOvertime, LoanAdvance, EmployeePerformance, CompanyRules
- EditPersonalInfo, EmergencyContact, Documents, DocumentViewer, BankDetails, AssignedAssets

3.3 Feature Details

A. Dashboard

- GET /employee/dashboard — managers, pending tasks, announcements, alerts, profile completion
- Quick clock + shortcuts to leave / expense / payslip / attendance

B. Attendance (dynamic rules)

- Rules: work start/end, grace, half-day cutoff, OT mins, weekend offs, holidays
- Clock-in !' present / late / half-day pending (HR queue)
- Clock-out late !' OT request for HR
- Corrections + WFH requests !' Manager or HR approve
- Off days block clock-in unless allowOffDayClockIn

C. Leave

- Create + list via POST/GET /employee/leave
- Manager OR HR can approve/reject
- GAP: cancel leave + leave balance APIs still TODO

D. Tasks

- List assigned tasks; PATCH status pending | in_progress | completed

E. Overtime / Expenses / Loans

- Submit OT with reviewing manager !' HR Approval Center
- Expenses with optional receipt !' HR review
- Loans & advances !' HR review

F. Payroll / Payslips

- List payslips + summary; open PDF from /employee/payroll/payslips/:id/pdf

G. Performance (dynamic)

- GET /employee/performance — manager reviews, KPIs, goals, task stats
- POST self-assessment attached to latest review
- Views ratings/feedback created in Manager Performance

H. Master Profile & Policies

- Personal, emergency, bank, documents, assets; completion %
- Avatar in Header + Drawer; company policies read-only

3.4 Employee APIs (summary)

```
GET /employee/dashboard | /managers | /tasks
PATCH /employee/tasks/:id/status
POST|GET /employee/leave | /overtime | /expenses
GET /employee/performance
```

POST /employee/performance/self-assessment
GET /employee/payroll/payslips...
POST|GET /employee/loans | /advances
GET|POST /employee/attendance/rules|clock-in|clock-out|corrections|wfh

3.5 Key Models

- Employee, User, ProfileCompletion, ManagerEmployeeAssignment
- Attendance, AttendanceRules, Correction, WfhRequest, HalfDayRequest, OvertimeRequest
- LeaveRequest, Task, ExpenseClaim, LoanAdvanceRequest
- Payslip, PayrollRun, SalaryStructure, PerformanceReview, Goal, CompanyPolicy

3.6 Dynamic vs Gaps

- DYNAMIC: dashboard, attendance, leave create/list, tasks, OT, expenses, payslips, loans, performance, policies, profile
- GAP: Inbox still demo; leave cancel/balance missing; notifications mostly local

4. Manager Module — Complete Detail

4.1 Purpose

Managers lead a team via ManagerEmployeeAssignment (primary/secondary). They assign tasks, approve leave/ WFH/corrections, write performance reviews, announce/message the team, and view live reports. Capabilities are gated by ManagerProfile.permissions (set by HR).

4.2 Navigation

Bottom Tabs

- Home — MHome | Team — MTeam | Approvals — MApproval
- Reports — MReports | Profile

Drawer / Stack

- Tasks (MTasks), Performance (MPerformance), Communication (MCommunicate)
- EmployeeDetail opened from Team member tap

4.3 Feature Details

A. Dashboard

- Live teamSize, presentToday (attendance snapshot), pendingApprovals, openTasks, permissions

B. Team

- List/search team; add from available pool; remove member
- Open full read-only profile via GET /manager/team/:employeeId
- Permission: teamManagement

C. Tasks

- Create/update/delete tasks for team members (deadline, priority, status)
- Permission: tasks

D. Approvals

- Queues: leave, WFH, correction — approve/reject with remarks
- Manager can fully decide Leave (same as HR for leave)
- Does NOT approve expenses, loans, half-day, OT (HR only)
- Permission: approvals

E. Performance

- Create review: employee, period, rating 1–5, feedback, KPIs
- Saved as completed; employee notified; appears on Employee Performance
- Permission: performance

F. Communication

- Team announcements + direct messages
- Permission: communication

G. Reports & Analytics (live)

- Overview: present/late/absent/on-leave today + pending + open tasks
- Attendance: present, late, half-day, WFH, leave, absent, clocked-in, on-duty, rate
- Productivity: tasks completed / in-progress / pending / rate
- Performance: reviews count + average rating
- Export PDF button removed
- Permission: reports

4.4 Manager APIs

```
GET /manager/dashboard
GET|POST|DELETE /manager/team... - teamManagement
GET /manager/employees/available
GET|POST|PATCH|DELETE /manager/tasks... - tasks
GET|PATCH /manager/approvals... - approvals
GET|POST /manager/reviews... - performance
GET /manager/reviews/:id/appraisal
GET|POST /manager/announcements | /messages - communication
GET /manager/reports/overview|attendance|productivity|performance
```

4.5 Permissions

ManagerProfile.permissions booleans (default true): teamManagement, approvals, performance, tasks, reports, communication. HR edits via ManagerPermissions screen (GET/PUT /hr/managers/:id/permissions).

4.6 Models

- ManagerProfile, ManagerEmployeeAssignment, Task, PerformanceReview, Goal
- Announcement, Message, LeaveRequest, WfhRequest, AttendanceCorrection, Attendance

4.7 Gaps

- No Manager recruitment UI (jobs API readable only)
- Drawer Team Reports may open hybrid Reports.jsx; tab MReports is the live analytics screen

5. HR Module — Complete Detail

5.1 Purpose

HR operates the EMS control plane: employee lifecycle, manager accounts & permissions, recruitment (job !' shortlist !' interview !' hire), attendance rules/roster/approvals, payroll generation, company policies, assets, and multi-queue Approval Center. Admin shares /api/hr on backend (HR_ADMIN); Admin mobile tabs are mostly placeholder.

5.2 Navigation

Bottom Tabs

- Home | Employees | Attendance | Recruit | Payroll | Profile

Drawer

- Approval Center, Attendance Rules, Company Policies, Managers, Assign Assets, Add Employee

Stack

- HRApprovalCenter, HRAttendanceRules, HREmployeeAttendance
- HRManagers, CreateManager, EditManager, ManagerPermissions, ManagerMonitor
- AddEmployee, EmployeeDetail, HRAssets, CompanyRules
- PostJob, ApplicantsPipeline, ApplicationReview, ShortlistedCandidates, ScheduleInterview

5.3 Feature Details

A. Employees

- List/search/paginate; create User+Employee; activate/deactivate; soft-delete
- Detail tabs: Profile, Salary, Managers, Assets, Documents
- Hire from candidate via select or AddEmployee with applicationId

B. Managers

- Create account + bulk team; edit; transfer; deactivate; permissions; activity/report

C. Recruitment

- Post jobs; pipeline by stage; Mark Review / Shortlist / Reject
- Open CV via Linking; Schedule Interview; Select/Hire !' Employee

D. Attendance

- Rules (hours, grace, half-day, OT, weekends) + holidays CRUD
- Live roster of real employees for today
- Employee history; half-day & OT HR approvals

E. Approval Center

- Leave, WFH, Correction, Half-day, OT, Expenses, Loans, Advances
- Approve/reject with remarks

F. Payroll

- Generate run from salary + attendance/LWP + OT + expenses + loans + bonus/deductions
- Payslip PDF via pdfkit under /uploads/payslips

G. Policies & Assets

- Company policies CRUD (employees read-only)
- Assign/remove assets on employees

5.4 Key HR APIs

```
GET|POST|PATCH|DELETE /hr/employees...
GET|POST|PATCH... /hr/managers... | /hr/assignments
POST|GET|PATCH /jobs...
GET|PATCH|POST /applications... (review|shortlist|reject|schedule|select)
GET|PATCH /hr/attendance/rules | roster | holidays | half-day | overtime | wfh | corrections
GET|PATCH /hr/expenses | /hr/loans | /hr/advances
POST|GET /hr/payroll/generate | runs | payslips | bonus | deductions
CRUD /hr/company-policies
```

5.5 Models

- Employee, User, ManagerProfile, Job, Application

- AttendanceRules, Holiday, Attendance, HalfDayRequest, OvertimeRequest
- LeaveRequest, ExpenseClaim, LoanAdvanceRequest, Payslip, PayrollRun, CompanyPolicy

5.6 Gaps

- Orphan screens HRLeave/HRExpenses/HRPerformance not in App stack (flows covered elsewhere)
- Some client TODOs: dashboard/ops, leave calendar, mark-absent
- Admin UI placeholder despite /api/admin backend

6. Cross-Role Workflows

6.1 Hire Path

- Candidate applies !' HR shortlist !' interview !' Select/Hire (or manual Add Employee)
- Result: role becomes employee + Employee master record

6.2 Leave Path

- Employee submits !' Manager Approvals AND HR Approval Center
- Either Manager or HR can finalize approve/reject

6.3 WFH / Correction

- Employee submits !' Manager and/or HR review

6.4 Half-day / OT

- Auto from clock rules !' HR Approval Center only

6.5 Performance

- Manager creates review !' Employee Performance shows rating/feedback/KPIs
- Employee optional self-assessment on latest review

6.6 Payroll

- HR generate run !' Employee Payslips + PDF

6.7 Expense / Loan / Advance

- Employee submits !' HR review !' payroll impact where applicable

7. Known Limitations & Future Work

- Employee Inbox still demo (wire communication inbox)
- Leave cancel + leave balance APIs missing
- No dedicated /hr/performance dashboard (manager-driven reviews exist)
- Admin tabs/audit UI not wired to /api/admin
- Approval Center documents filter empty
- Manager/HR analytics PDF export not implemented
- /hr/dashboard/ops may be incomplete

7.1 Suggested Test Checklist

- Candidate: signup !' setup !' apply with PDF !' withdraw/reschedule
- HR: shortlist/reject/open CV !' schedule !' hire
- Employee: clock-in/out !' leave !' expense !' payslip !' view manager review
- Manager: add team !' task !' approve leave !' create performance review !' reports
- HR: attendance rules + roster !' OT/half-day approve !' generate payroll

— End of Brilliance EMS Complete Role Documentation —

